

PAVULURI KUMAR

Boston, USA (Open to Relocate) | +1(203)-518-4105 | pavuluri.kumar@aogjob.com

SUMMARY

Data Engineer with 4 years of experience building scalable ETL pipelines, real-time data streaming solutions, and cloud-based architectures on AWS and Azure. Proficient in Python, PySpark, and SQL, with expertise in big data processing, machine learning integration, and data modeling. Designed and optimized workflows, automated deployments with CI/CD, and implemented data governance to maintain compliance and data integrity. Experienced in handling large datasets, improving query performance, and ensuring efficient data processing.

PROFESSIONAL EXPERIENCE

Cigna

Data Engineer

Remote, USA

September 2023 - Present

- Reduced ETL pipeline processing time by 40% by developing and managing Apache Airflow ETL pipelines using Python and Apache Spark, optimizing schema validation and data quality standards.
- Facilitated CI/CD automation in AWS through Terraform and GitHub Actions, executing a data migration to improved AWS Redshift configuration and delivering annual cost efficiencies of around \$150K/yr.
- Devised AWS CloudWatch Events with AWS lambda to trigger AWS Glue jobs, automating the processing of 18 M records daily and ensuring real-time updates to AWS Redshift, reducing manual interventions and operational costs.
- Modernized data modeling by implementing Data Vault, enhancing flexibility for schema changes and reducing query times by 30%, facilitating real-time analytics and faster decision-making for regulatory and business needs.
- Implemented Change Data Capture (CDC), real-time synchronization of EDI transactions using Debezium and Kafka reducing data latency by 15% ensuring, enhancing data accuracy and consistency.
- Collaborated cross-functionally with technical teams and business stakeholders and to enforce data governance policies, ensuring compliance with HIPAA, X12 and HL7 standards.

Sigma InfoSolutions

Data Engineer

Bengaluru, India

April 2019 - July 2021

- Utilized Azure Data Factory ETL for integration and processing of 12M records daily from on-premises CRM systems and Cassandra NoSQL Finance Databases, alongside Azure Databricks for data profiling and consistency.
- Managed data warehousing solutions in Azure Synapse Analytics, supporting over 80+ real-time reports using Power BI DAX, while ensuring reliable data flow for customer-focused analytics and reporting across business units.
- Fine-tuned ML model training, experimentation, and A/B testing in Azure Databricks, utilizing Azure ML APIs to forecast customer behavior, which reduced churn by 15%.
- Created Python scripts to analyze resource usage for long-running jobs within Azure Databricks to optimize cluster performance and increase availability, resulting in a reduction of running time by 5 hours/day.
- Developed CI/CD pipelines in Azure DevOps for continuous deployment of ETL processes and integration with Databricks MLOps for model delivery, accelerating deployment times by 35% and reducing manual errors.
- Optimized SQL queries and Database performance to enhance query performance, streamline data retrieval, and reduce average processing time by around 20%.

PROJECTS

Live Data Stream Processing System

- Devised data streaming solution to handle dataset of 10 million records, utilizing TCP/IP Socket for data acquisition, Spark Streaming for real-time processing, Kafka for message queuing, and Elasticsearch for indexing and querying.
- Enhanced pipeline, integrating sentiment analysis via OpenAI LLM for real-time analysis of customer feedback and optimizing streaming efficiency to increase average throughput to 160 records per second.

AWS Reddit Data Workflow

- Accomplished data pipeline workflow using Apache Airflow, orchestrating ETL processes and integrating the Reddit API for data ingestion, with Docker employed for Airflow environment setup and initialization
- Employed AWS Glue with PySpark for data transformation and cleaning, integrating with services (S3, Redshift, and Athena), achieving 60% improvement in query response times by migrating processing tasks from Athena to Redshift.

EDUCATION

Masters of Science in Computer Engineering

San Jose State University

May 2023

San Jose, CA

Bachelor of Engineering

Birla Institute of Technology

May 2019

Ranchi, India

TECHNICAL SKILLS

Languages: Python, SQL, Linux Scripting, Java

Cloud & Big Data Technologies: AWS S3, Glue, Lambda, Redshift, CloudWatch, IAM, EC2, Databricks, Azure Synapse Analytics, Data Factory, ADLS, Snowflake, Terraform, Apache Airflow, Apache Spark, Apache Kafka, Kubernetes, Hadoop

Python Packages & Other Tools: PySpark, Boto3, psycopg2, Pandas, NumPy, Scikit-Learn, PyTorch, Advanced Excel, Docker, Jenkins, Jira, Azure Devops, Informatica, Qlik Sense, Data Build Tool, Tableau, Power BI, Github, Control-M

CERTIFICATION

AWS Certified Data Engineer Associate